My recommendation is to rebuild this as a rigorous study of cross-lingual lexical representations, with Bangla screening-item design as the motivation. That is the strongest route using your existing work. If you want to keep an application-heavy framing for ESWA, you will also need an independently evaluated item-selection system.

Better wording alone will not address this rejection. The main work is stronger controls, independent evaluation, and narrower claims. No revision can guarantee acceptance.

1. Decide what the paper will actually contribute

Your current manuscript tries to establish phonological knowledge, lexical transfer, tokenizer causality, localization, and screening usefulness simultaneously. The evidence is uneven across those claims.

Make the central question:

> Do multilingual model representations distinguish Bangla words from plausible pseudowords beyond orthographic and lexical surface cues, when the classifier is trained only on English?

Then investigate tokenization and localization as secondary explanations.

A possible title is:

> *Beyond surface cues: Evaluating English-to-Bangla lexical transfer in multilingual language models*

Keep dyslexia in the motivation and discussion unless you add application validation. You do not need to invent a new neural architecture, but you do need a contribution that goes beyond ranking five models.

2. Audit the existing results before generating more

First, ensure that every claim, table, and figure can be regenerated consistently.

Issue in the current PDF	Required correction
Some Table 4 gains do not equal peak minus layer-0 scores—for example, mT5 English shows 0.65 → 0.90 but a gain of +0.23.	Recompute from saved predictions and generate tables automatically.
Figure 6 appears inconsistent with its narrative about which models show ablation effects.	Trace the exact experiment, task, layer, and dataset behind the figure; replace or explain it.
Section 4.6 describes specificity with the opposite subtraction order from the positive-specificity interpretation.	Define one equation and use it consistently.

English-only layer selection is stated, but several results emphasize peak Bangla performance.	Separate English-selected confirmatory results from exploratory Bangla layer curves.
RSA summaries include descriptions such as “high” instead of numerical results.	Provide actual values, sample sizes, and uncertainty.

These are not merely presentation problems: they affect whether readers can trust the experimental conclusions.

3. Build a larger, harder, independently checked lexical dataset

Expanding all seven categories equally is unnecessary. Prioritize the task that supports the paper’s main contribution.

Construct real-word/pseudoword pairs matched on more than character length:

- Unicode code-point length and grapheme-cluster length.
- Syllable count and consonant clusters.
- Bangla conjunct and vowel-sign patterns.
- Character n-gram probability.
- Orthographic neighborhood: how similar an item is to existing words.
- Frequency bands for the real words.

Use multiple pseudoword-construction procedures. Otherwise, a model may recognize the generator’s habits rather than lexical status.

Have at least two independent native-language annotators check lexical status and plausibility, ideally including someone with relevant linguistic expertise. Record disagreement and adjudication. Document the sources and construction rules.

A practical planning target is several hundred independently verified pairs per language, subject to annotation capacity and a precision analysis. That is a proposed target, not a publication threshold.

Reserve a new Bangla test set before further model development. The existing 50-item evaluation has already informed your conclusions; treat it as exploratory evidence.

4. Strengthen the baselines—the current five surface features are insufficient

Beating length-related counts does not establish that models have learned phonology. They may exploit richer spelling patterns.

Add these comparisons:

Baseline	What it tests
First-character and first-grapheme classifiers	Whether onset identity explains the labels.

Character n-gram classifier	Whether local spelling patterns explain performance.
Character language-model score	Whether orthographic probability separates words and pseudowords.
Lexical frequency and neighborhood features	Whether familiarity and similarity explain performance.
Combined surface-feature model	Whether an LLM contributes beyond a stronger conventional baseline.
Surface features plus LLM representations	Whether representations provide incremental value.

Keep the transfer conditions explicit. A baseline trained using labeled Bangla examples is a supervised Bangla reference, not a zero-shot competitor. Likewise, identify any external Bangla lexicon or unlabeled corpus it uses.

The key experiment is whether LLM features improve a strong baseline on the untouched test set. If they do not, report that outcome; a well-supported negative result can still be informative.

5. Make the evaluation resistant to leakage and selection bias

Use this sequence:

1. Group related items—minimal pairs, shared roots, and generated variants—so related examples cannot cross training and validation splits.
2. Select layers, regularization, and preprocessing using English training/validation data only.
3. Freeze the complete procedure.
4. Evaluate on the untouched Bangla set.
5. Report paired differences between models and baselines, with confidence intervals.

Bootstrap at the level of independent item families when examples are related. Separately evaluate variation across training splits and seeds; bootstrapping fixed predictions does not measure all training uncertainty.

Add label-randomization checks and appropriately designed control tasks. Control-task selectivity has an established methodological basis, so use it to strengthen validity rather than present it as your invention. Hewitt and Liang, 2019

Choose a small number of primary comparisons in advance. Label the remaining layer, model, and subgroup analyses exploratory.

6. Fix the tokenization claim without committing to an expensive pretraining project

The practical option is to change the claim to:

ByT5 outperforms mT5 under this evaluation; tokenization is a plausible contributor, alongside architectural and training differences.

The original ByT5 paper describes changes beyond replacing the tokenizer, including changes in model dimensions and encoder–decoder allocation. Your comparison therefore needs a fuller confound discussion than encoder depth alone. ByT5 paper

Then add evidence that tests the proposed explanation:

- Measure token fragmentation for each item.
- Evaluate performance across fragmentation and orthographic-complexity groups.
- Compare more than one size within the mT5 and ByT5 families, if feasible.
- Test robustness to linguistically justified spelling variation.
- Measure inference cost as well as accuracy.

These analyses can support a mechanism hypothesis, but do not isolate tokenizer causality.

A genuinely controlled causal experiment would require separately training comparable models while controlling data, architecture, objective, and training budget. **I would not make that your first priority.**

7. Repair—or reduce—the localization contribution

For stronger localization evidence:

- Compare top-ranked and random ablations using the **same number of units**.
- Test several ablation sizes and report performance-drop curves.
- Include proportional ablations because hidden widths differ.
- Select units using English training data only.
- Distinguish direct representation ablation from interventions propagated through the model.
- Test whether findings repeat across splits and an independent dataset.

Describe a null effect as “limited sensitivity to this intervention,” rather than proof of distributed storage.

For mT5, avoid describing the affected features as a useful transferable signal when its intact classifier does not reliably beat the surface baseline.

If localization remains inconsistent, remove it from the main contribution. A focused, reliable paper is stronger than a broader paper with unresolved mechanisms.

8. Add independent validation that matches your intended publication route

You have two realistic routes:

Route	Additional evidence needed

Empirical NLP / representation study	Independent Bangla test data, stronger baselines, robust evaluation, and ideally replication using another dataset or another language with qualified annotators.
Applied intelligent-system study	An actual item-selection or quality-control component, compared with conventional methods and independently evaluated by relevant experts.

For the application route, a feasible experiment is:

> Given a pool of candidate Bangla lexical-decision items, does an LLM-assisted selection method produce more valid, plausible, appropriately categorized items than surface rules alone?

Compare surface-only selection, LLM-only selection, and a combined method. Have experts assess the outputs without knowing which method selected them. Define outcomes beforehand—for example, valid-item rate, category agreement, and review time.

This would demonstrate **item-construction utility**. It would still not demonstrate dyslexia-screening accuracy. Claims about detecting dyslexia require a separate participant-based validation study.

9. Rewrite and release the paper around what survives

Once the new experiments are complete:

- Rewrite the abstract around the strongest confirmed result.
- Remove claims that “lexical” necessarily means “phonological.”
- Avoid attributing differences to parameter count, coverage, or objective when those factors were not independently controlled.
- Remove unsupported compute-cost comparisons and deployment estimates.
- Release the dataset where permitted, annotation guidance, split identifiers, code, configurations, and per-item predictions.
- Explain explicitly what is new relative to the closest prior studies.

Your contribution should become something like:

> An independently annotated Bangla lexical benchmark and a reproducible evaluation showing when English-trained multilingual representations add information beyond strong orthographic baselines.

That is a **candidate contribution**, contingent on the experiments and a fuller novelty check.

The order I would follow is:

1. Audit and reconcile current results.
2. Specify the main hypothesis and stronger baselines.
3. Construct and independently annotate new data.
4. Freeze the evaluation protocol.

5. Run the primary transfer experiment.
6. Add robustness and replication.
7. Retain localization only if it holds up.
8. Add expert-evaluated item selection if pursuing the application route.
9. Rewrite and select a venue suited to the resulting evidence.

Your most valuable next experiment is not a sixth model. It is an untouched Bangla test set evaluated against a strong orthographic baseline. That result will tell you which claims the revised paper can credibly make.